

Machine Learning-Based Traffic Congestion Prediction for Smart Urban Transportation Systems

Mohammad Shihab Mustafa, Taehyun Kim, Luiza Harutyunyan
Pacific States University
Los Angeles, California, USA

Abstract—Traffic congestion is a major transportation problem in Los Angeles and other large urban areas. This study uses smart city transportation data to analyze congestion patterns and test machine learning methods for prediction. The dataset includes traffic volume, directional flow, time-related variables, and location-based information.

The data was cleaned, transformed, and used to create features such as congestion level and traffic flow indicators. Exploratory analysis showed clear weekday and weekend differences in traffic volume. Machine learning experiments were then conducted using regression and classification models. The results showed strong prediction performance, including high accuracy in classifying congestion into low, medium, and high categories.

The study shows that traffic data can support congestion monitoring and transportation planning. The findings may help future smart city systems improve traffic analysis, mobility planning, and data-driven decision-making.

Index Terms—Los Angeles, machine learning, smart cities, urban transportation, traffic congestion, data analytics, predictive modeling.

I. INTRODUCTION

Most big cities struggle with too many cars on the road, yet none more so than Los Angeles [1], [4]. Vehicles fill highways each day because people drive far to work while homes and jobs pack tightly together. Delays pile up - drivers wait longer, buses fall behind schedule, companies lose time, emissions rise. When traffic crawls, engines burn extra gas even when barely moving forward. Productivity dips where roads jam, just as smog builds from stopped and starting again.

Now cities use smart tech to track how traffic moves, mixing live updates with past records [2], [11]. Traffic jams show up clearer when scholars combine bus and train details with air quality numbers, linked through clever software tricks. Before backups grow bad, forecast systems spot likely choke points by spotting hidden trends in the noise. Not just explaining pileups once they're done, these number-crunching methods guess where slowdowns might pop up next [3], [5].

The purpose of this research is to analyze traffic congestion patterns in Los Angeles using smart city data. By improving congestion prediction through machine learning and smart transportation analytics, this research may support intelligent urban infrastructure planning, reduce transportation inefficiencies, and contribute to smarter mobility systems across major U.S. cities.

II. LITERATURE REVIEW

Previous traffic prediction studies mainly focused on historical traffic speed, traffic flow, and sensor-based transportation data. Traditional statistical methods were commonly used to estimate congestion patterns and traffic movement across urban road networks.

Recent research has increasingly applied machine learning and deep learning techniques to traffic forecasting problems. Methods such as Linear Regression, Random Forest, LSTM networks, and spatiotemporal neural networks have been used to analyze traffic flow behavior and predict congestion patterns using transportation datasets.

Several studies have also examined how environmental and temporal variables influence traffic conditions. Weather conditions such as rain, fog, and low visibility may affect driving behavior and traffic density. In addition, traffic patterns often vary between weekdays and weekends as well as across different time periods of the day.

Existing research suggests that traffic prediction performance may improve when transportation, environmental, spatial, and temporal variables are analyzed together rather than individually [1], [4], [11].

III. RESEARCH QUESTION AND OBJECTIVES

The primary research question of this study is:

Can traffic congestion levels in Los Angeles be predicted using traffic volume, environmental conditions, and temporal factors?

The supporting objectives are:

- Identify congestion hotspots in Los Angeles.
- Analyze peak-hour and off-peak traffic patterns.
- Examine relationships between traffic and environmental variables.
- Compare machine learning models for congestion prediction.
- Provide visual insights that can support transportation planning.

IV. CONCEPTUAL FRAMEWORK

The conceptual framework for this study is shown in Fig. 1. Transportation variables, environmental variables, and temporal-spatial factors are used to analyze and predict traffic congestion.

Transportation variables such as traffic volume and average speed directly describe traffic flow. Environmental variables

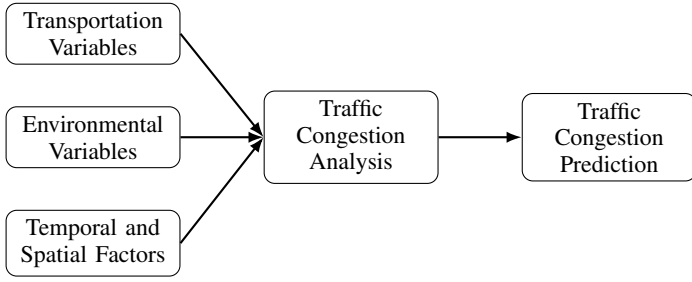


Fig. 1. Conceptual framework for traffic congestion prediction.

such as weather, visibility, and air quality may affect driver behavior. Temporal and spatial factors such as hour of day, day of week, and location provide additional context for understanding congestion patterns.

V. DATASET DESCRIPTION

The dataset for this research will be collected from public smart city data sources, including the Los Angeles Open Data portal, OpenAQ, and NOAA climate data. These sources can provide transportation, air quality, weather, time, and location-based variables.

Each observation represents a traffic condition at a specific time and location. This makes the dataset useful for studying traffic flow patterns across different hours, days, and areas of Los Angeles.

TABLE I
EXAMPLE DATASET VARIABLES

Variable	Description
Timestamp	Date and time of traffic observation
Location	Street name, sensor location, or geographic coordinates
Traffic Volume	Number of vehicles detected during an interval
Average Speed	Mean vehicle speed during the interval
AQI	Air Quality Index near the traffic observation
Weather Condition	Weather status such as clear, rain, fog, or cloudy
Day of Week	Weekday or weekend indicator
Hour	Hour of the day based on timestamp
Congestion Level	Target variable such as low, medium, or high congestion

VI. METHODOLOGY

This study follows a structured methodology for traffic congestion analysis and prediction using smart city transportation data. The workflow includes data collection, preprocessing, feature engineering, exploratory data analysis, predictive modeling, and model evaluation. The purpose of the methodology is to transform raw transportation data into meaningful analytical insights that support congestion prediction and traffic analysis.

A. Data Collection

Traffic-related variables will be collected from Los Angeles public transportation datasets. Weather and air quality variables may be added from public environmental data sources. These datasets will be combined by matching records based on time and location when possible.

B. Data Preprocessing

Raw traffic datasets may contain missing values, duplicate observations, inconsistent timestamps, and unrealistic sensor readings. Missing numerical values such as traffic speed or traffic volume may be handled using interpolation, mean imputation, or median imputation. Records with excessive missing information may be removed.

Noise filtering will also be applied to remove unrealistic values, such as extremely high vehicle speeds or sudden traffic spikes caused by faulty sensors. Timestamp formatting will be standardized so that hour, day of week, and weekend indicators can be created.

C. Feature Engineering

Feature engineering will be used to create variables that better describe congestion behavior. Possible engineered features include congestion index, traffic density, speed-to-volume ratio, peak-hour indicator, weekend indicator, and environmental impact variables.

For example, a simple congestion index can be calculated as:

$$Congestion\ Index = \frac{Traffic\ Volume}{Average\ Speed + 1we} \quad (1)$$

A higher value may indicate heavier congestion because traffic volume is high while average speed is low.

VII. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to examine traffic patterns in the Los Angeles traffic dataset before predictive modeling. The purpose of this stage is to identify traffic trends, compare traffic behavior across different days, and support later model development using real dataset observations.

A. Traffic Volume by Day of Week

Fig. 2 presents the average traffic volume observed across different days of the week. The results indicate that traffic volume is generally higher during weekdays, particularly on Wednesday and Friday, while weekend traffic volume is noticeably lower. These results indicate that weekday commuting activity contributes significantly to traffic congestion patterns in Los Angeles.

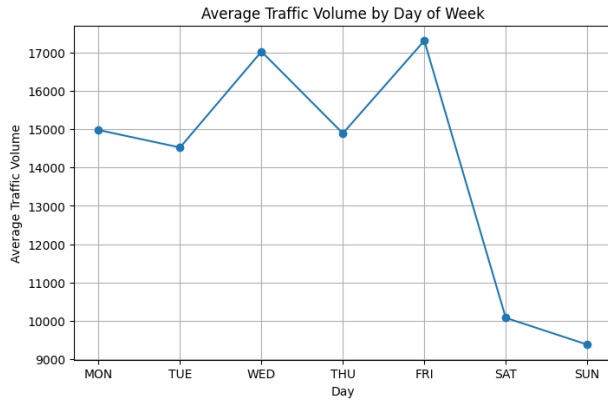


Fig. 2. Average traffic volume by day of week

B. Correlation Matrix Analysis

Fig. 3 presents the correlation matrix of major traffic variables used in this study. The analysis shows that total traffic volume has strong positive relationships with directional traffic counts, while congestion index values also demonstrate strong relationships with overall traffic volume. Correlation analysis helps identify relationships among variables and supports feature selection for predictive modeling.

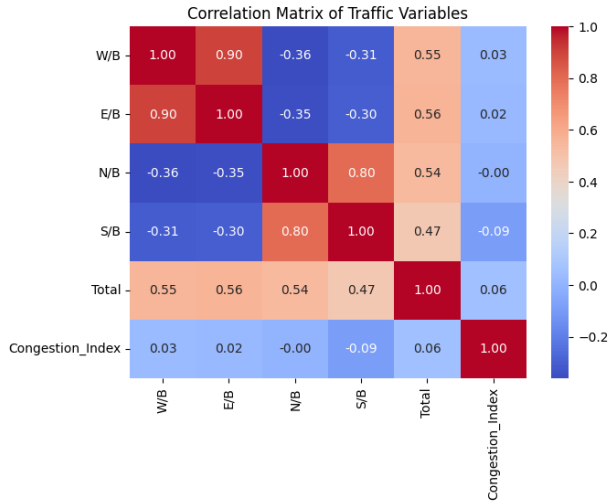


Fig. 3. Correlation matrix of traffic variables

VIII. MODELING APPROACH

Traffic congestion can be predicted using both regression and classification models.

A. Regression Models

Regression techniques apply when traffic congestion shows up as numerical data, such as indices or average speeds. Simple in design, Linear Regression often comes first in discussion. Without requiring straight-line relationships, Random Forest manages intricate trends effectively. Step by step, Gradient Boosting improves forecasts by adjusting earlier errors.

B. Classification Models

Classification models are used to categorize traffic congestion into predefined classes such as low, medium, or high congestion. Common machine learning approaches for classification include Logistic Regression, Decision Trees, Random Forest Classifier, and Support Vector Machines.

These models help identify congestion patterns based on traffic flow variables and engineered transportation features. Among these approaches, Random Forest classification is particularly useful because it can capture nonlinear relationships and improve predictive performance across complex traffic datasets.

C. Time-Series Forecasting

Time-series forecasting methods may be used to predict future traffic congestion patterns based on historical transportation data. Common forecasting approaches include ARIMA models, Long Short-Term Memory (LSTM) networks, and other deep learning techniques designed for sequential data analysis.

These models can help identify temporal traffic patterns across different hours, days, and traffic conditions. Time-series forecasting may improve the ability of smart transportation systems to anticipate congestion trends before severe traffic buildup occurs [1], [2], [9].

IX. PROPOSED SYSTEM ARCHITECTURE

The proposed system architecture is shown in Fig. 4. It explains how raw traffic and environmental data can be transformed into useful congestion predictions.

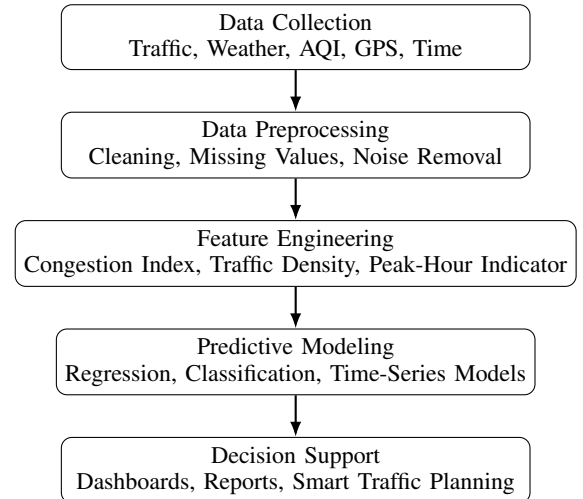


Fig. 4. Proposed system architecture for traffic congestion prediction.

The first layer collects transportation and environmental data. The second layer cleans and prepares the dataset. The third layer creates new features that better represent congestion behavior. The fourth layer trains predictive models. The final layer presents results through dashboards, reports, and decision-support systems.

X. EVALUATION METHODS

The predictive models will be evaluated using metrics that match the modeling task.

A. Regression Evaluation Metrics

Regression models may be evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score. MAE and RMSE measure prediction error, while R^2 explains how much variation in congestion is captured by the model.

B. Classification Evaluation Metrics

Classification models may be evaluated using accuracy, precision, recall, and F1-score. These metrics show how well the model classifies congestion into low, medium, and high categories.

XI. EXPERIMENTAL RESULTS

Before any models learned patterns, raw data passed through cleaning steps like fixing gaps, changing formats, creating new variables, removing bad timestamps. Experiments on Los Angeles road flow tested how well the suggested method forecasts gridlock. Instead of skipping errors, each anomaly got reviewed early. Training only began once inputs met quality thresholds.

A Linear Regression model was first developed to predict total traffic volume using directional traffic counts. The regression model achieved a Mean Absolute Error (MAE) of approximately 718 vehicles and an R^2 score of 0.986, indicating strong predictive performance.

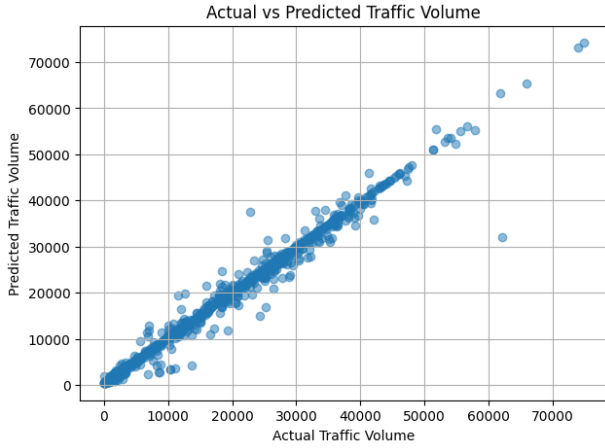


Fig. 5. Actual versus predicted traffic volume using Linear Regression

Running two separate tests with Random Forest classifiers helped assess how well congestion could be predicted. One setup included both direction-specific vehicle numbers and overall traffic amount. In contrast, the other left out the total volume to see if movement trends by direction were enough on their own. Whether full volume data improved accuracy was the key question behind the comparison.

The first model achieved an overall accuracy of approximately 99.24%, while the second model achieved approximately 98.40% accuracy using only directional traffic variables. These results indicate that directional traffic patterns alone provide strong predictive capability for congestion classification in Los Angeles.

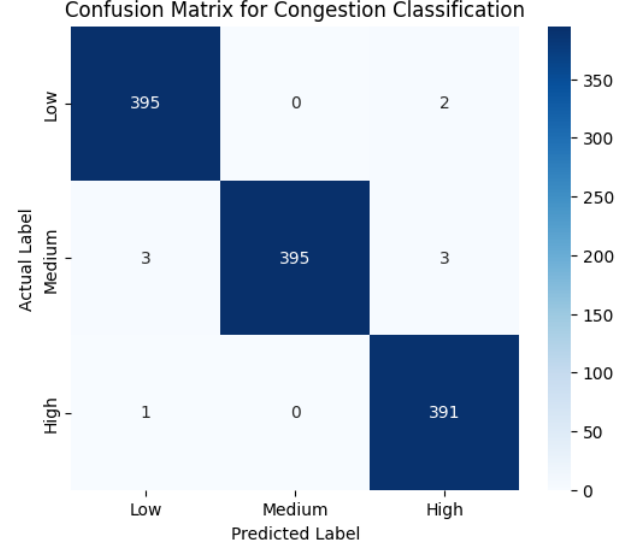


Fig. 6. Confusion matrix for congestion classification

Feature importance analysis was also conducted using the Random Forest model. The results show that total traffic volume contributed most strongly to congestion classification, while directional traffic counts also supported prediction performance.

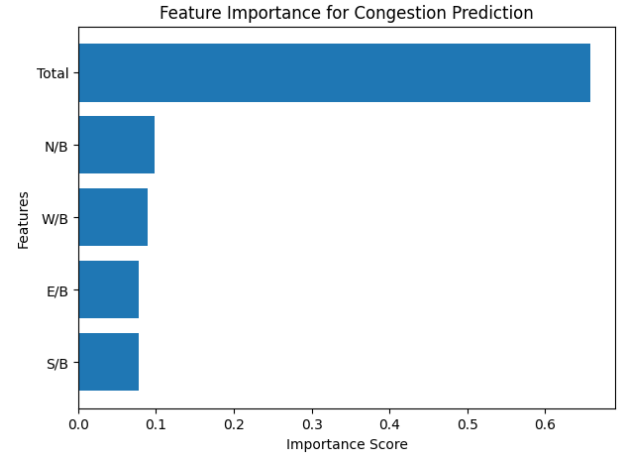


Fig. 7. Feature importance for traffic congestion prediction

Overall, the experimental results show that machine learning models can effectively analyze and predict traffic congestion patterns using smart city transportation data.

XII. EXPECTED INSIGHTS

The findings of this study are expected to identify major traffic congestion trends across different days, locations, and traffic conditions within Los Angeles. The analysis may also help determine how weekday and weekend traffic patterns differ across urban transportation networks.

The experimental results suggest that directional traffic flow patterns provide strong predictive capability for congestion classification. In addition, temporal variables such as day of the week and traffic timing may improve the performance of congestion prediction models.

The broader implications of this research may extend beyond Los Angeles by supporting data-driven transportation planning in rapidly growing urban areas across the United States. Improved congestion forecasting systems may help transportation planners reduce traffic-related delays, improve commuter mobility, optimize transportation infrastructure, and support sustainable urban development.

The results of this study may also support future smart transportation systems that rely on machine learning, predictive analytics, and intelligent traffic management technologies.

XIII. LIMITATIONS

This study faces several limitations. One, available urban data might lack entries, show gaps, or present contradictions due to faulty sensors or uneven gathering methods. Such flaws could influence how well models perform.

Unexpected incidents - like crashes, storms, or big public gatherings - can shift how vehicles move on roads. When floods block highways or a stadium empties suddenly, patterns change fast. Such moments often slip through data collection. Construction zones pop up without warning, altering usual routes. Even parades or emergencies reshape flow in ways models might miss. Information gaps mean some disruptions stay invisible in datasets.

Another limitation of this research is its focus on Los Angeles. Because urban layouts differ - roads, transit networks, how people travel - the findings might not fit elsewhere exactly. Traffic patterns observed in Los Angeles may differ under other urban conditions.

Responsible handling of transportation information matters, especially within smart cities. Though data may come from public sources, care must go into protecting individual privacy. Transparency becomes key - people deserve clarity about how their movements are tracked. Fairness cannot take a back seat either; decisions shaped by such data ought to serve everyone equally. Thoughtful practices help maintain trust, even when systems run on automation.

XIV. ETHICAL CONSIDERATIONS

Even if the data comes from open smart city sources, privacy questions remain. Movement records tied to roads or places might expose trends in how people travel across areas. As a result, personal details must stay out of analysis - only summaries of vehicle flow matter here. Insights come from group behaviors, never single users.

Fairness remains an important consideration in smart city transportation systems. When cities rely on traffic forecasts to shape development, benefits ought not cluster around wealthy districts or business zones alone. Instead, neighborhoods relying on buses or subways deserve attention - especially those already burdened by dirty air. Equity means progress spreads evenly: safer streets, easier travel, real access, everywhere. Transportation improvements should benefit communities fairly and support equitable urban mobility outcomes.

Understanding how models work matters just as much as their outcomes. When transportation planners rely on machine learning forecasts, knowing what drives those results becomes essential. Tools that improve AI transparency and highlight key predictive factors may improve model interpretability. Such approaches let users see inside the system, making choices less dependent on opaque algorithms alone.

XV. FUTURE SCOPE

Later studies might build on this work through live traffic inputs drawn from intelligent detectors, visual monitoring tools, vehicle tracking units, and networked automobiles. With instant updates fed into forecasting models, gridlock estimation can adapt swiftly alongside shifting city movement patterns [8], [14]. When applied broadly, such methods may ease blockages nationwide, supporting smoother travel flows, improved resource use, and longer-term metropolitan planning goals.

Future work could explore deep learning methods like LSTM, GRU, Transformers, or GNNs. With time, these systems show potential in modeling intricate traffic behaviors across space and duration. Unlike older techniques, they handle sequence data through adaptive internal states. Their structure allows processing of dynamic urban flows with higher precision [2], [6].

Later research might look into smart control of traffic lights by applying reinforcement learning methods. Rather than merely forecasting congestion, upcoming models could suggest dynamic changes to light timing so that wait times drop while movement through roads becomes smoother [12].

Future studies might examine Los Angeles traffic alongside that of other large cities, testing if forecasting methods work reliably in varied urban settings. By looking across metropolitan areas, researchers could better understand how road networks and travel habits affect congestion forecasts [7].

It might be that using explainable artificial intelligence improves clarity within traffic congestion forecasts. Depending on the approach, insights into key influencing factors can become more accessible to urban mobility analysts [13].

Intelligent transportation forecasting systems could play a role in updating urban infrastructure across major U.S. cities. With time, such tools might align closely with broader improvements in how metropolitan areas manage mobility. Their integration may follow patterns seen in technology-driven upgrades elsewhere. Yet their value will depend on coordination between planners and local conditions. Not every region will adopt them at the same pace. Still, potential

exists where data-informed decisions are prioritized. Over years, these systems may become embedded in routine operations. Where implemented carefully, benefits could emerge gradually. Progress, however, remains tied to investment and governance structures.

XVI. PRACTICAL IMPLEMENTATION PLAN

A practical implementation of this framework may involve collecting transportation, environmental, and temporal data from public smart city data sources. After data collection, preprocessing techniques such as missing value handling, timestamp standardization, and feature engineering may be applied to prepare the dataset for machine learning analysis.

Machine learning models including Linear Regression, Random Forest, and time-series forecasting methods may then be trained to analyze traffic congestion behavior and predict future traffic conditions. Model performance can be evaluated using regression and classification metrics such as MAE, RMSE, accuracy, precision, recall, and F1-score.

The resulting prediction framework may support dashboard-based traffic monitoring systems capable of identifying congestion hotspots, high-traffic periods, and transportation trends across urban road networks.

Such systems may support transportation agencies, smart city planners, and urban infrastructure departments in making data-driven decisions for congestion mitigation and mobility improvement initiatives.

XVII. CONCLUSION

This study explored the use of smart city transportation data and machine learning techniques for traffic congestion analysis in Los Angeles. Traffic-related variables, directional flow patterns, temporal features, and environmental conditions were examined to support congestion prediction and transportation analysis.

Experimental results demonstrated that machine learning models can effectively predict traffic congestion patterns using real transportation data. The regression and classification experiments produced strong predictive performance, while comparative analysis showed that directional traffic flow patterns alone provide high congestion classification accuracy even without total traffic volume information.

The study also demonstrated the importance of preprocessing, feature engineering, exploratory analysis, and model evaluation within intelligent transportation research. The findings suggest that temporal traffic patterns and directional traffic behavior contribute significantly to congestion forecasting performance.

Although limitations such as incomplete data, unexpected road events, and location-specific traffic behavior remain, this research provides a practical framework for future smart transportation studies. Future extensions may include real-time traffic integration, environmental forecasting variables, explainable artificial intelligence techniques, and intelligent traffic signal optimization.

If expanded further, this framework may contribute to the development of intelligent transportation systems that support safer, more efficient, and more sustainable urban mobility solutions throughout the United States.

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